Cybersecurity Incident Report

Section 1:
The website may be experiencing a TCP SYN flood attack, a type of denial-of-service (DoS) attack. The attacker sends repeated TCP SYN packets to the web server, potentially exhausting its resources and preventing legitimate users from establishing connections. The logs show that: The packet capture shows repeated TCP SYN packets from source IP address 203.0.113.0 to destination IP address 192.0.2.1 on port 443. The repeated requests continue throughout the capture. The logs also show server responses with SYN-ACK and RST, ACK flags, as well as an HTTP 504 Gateway Time-out response. This event could be:A TCP SYN flood attack causing a denial of service. The repeated SYN packets and connection failures are consistent with this type of attack. Additional evidence would be needed to determine whether the source IP address is spoofed or whether the attack involves multiple sources.

Section 2:
The three steps of the handshake: 1. SYN: The client sends a SYN packet to request a TCP connection. 2. SYN-ACK: The server responds with a SYN-ACK packet to acknowledge the request. 3. ACK: The client sends an ACK packet to complete the connection. Explain what happens when a malicious actor sends a large number of SYN packets all at once: The attacker sends repeated SYN packets to the web server. The server responds with SYN-ACK packets and may reserve resources for pending connections while waiting for the final ACK. If the connections remain incomplete, the server's connection backlog and resources may become exhausted, preventing legitimate clients from connecting. Explain what the logs indicate and how that affects the server:

The packet capture shows repeated SYN packets from 203.0.113.0 to 192.0.2.1 on port 443, without evidence of a corresponding completed handshake for each repeated request. The server also sends reset packets, and an HTTP 504 Gateway Time-out response appears in the capture. These patterns are consistent with a TCP SYN flood that may be overwhelming the server's ability to handle connection requests.

As a result, legitimate employees and website visitors may experience slow loading times, connection timeouts, or failed access to the sales page. This disruption can interfere with employees' work, reduce customer satisfaction, and potentially cause lost sales.

Ways to secure the network and prevent future attacks:

The organization should implement SYN flood protection, such as SYN cookies, configure appropriate firewall rate limits, monitor unusual connection patterns, and consider a DDoS protection service. Blocking one IP address alone may not be sufficient if the attacker changes or spoofs source addresses.